

VINAY REDDY NOMULA

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Professional Summary

Data Engineer with 4 years of expertise in **AWS cloud architectures, real-time data processing, and scalable infrastructure automation** across finance, healthcare, and retail. Skilled in **Terraform, CI/CD workflows, AWS IAM security controls**, and optimizing **ETL pipelines for high-performance analytics**. Proven ability to enhance **data reliability, reduce latency, and ensure compliance with PCI-DSS & HIPAA**. Strong experience in **Apache Kafka, Spark, AWS (S3, Redshift, Glue, Step Functions), Python, SQL, and Power BI**, supporting business decision-making through actionable insights.

Technical Skills

- **Programming & Scripting:** Python, SQL, Scala, Shell Scripting
- **Big Data & Processing:** Apache Spark, Hadoop, Hive, Pig
- **Streaming & Real-Time Processing:** Apache Kafka, Spark Streaming
- **ETL & Workflow Orchestration:** Apache Airflow, AWS Glue, Talend
- **Databases & Warehousing:** AWS Redshift, Snowflake, PostgreSQL, MySQL, DynamoDB, MongoDB
- **Cloud Platforms:** AWS (S3, Redshift, Lambda, CloudWatch, Glue, IAM, Step Functions)
- **Infrastructure & DevOps:** Docker, Jenkins, Git, Terraform, CI/CD Pipelines, Kubernetes
- **Data Visualization & Reporting:** Power BI
- **Data Governance & Compliance:** Data Lineage, Quality Checks, PCI-DSS, HIPAA
- **Testing & API Development:** Pytest, Postman, Swagger UI
- **Version Control & Collaboration:** Git, Bitbucket, Azure DevOps
- **Monitoring & Debugging Tools:** Jupyter Notebook, Visual Studio Code, SQL Server Management Studio, Datadog

Professional Experience

American Express, Plano, TX

April 2024 – Present

Data Engineer

Responsibilities:

- **Developed real-time fraud detection pipelines** using **Apache Kafka, Spark Streaming, and AWS (S3, Redshift, Glue, IAM)**, leveraging **Terraform** for **automated infrastructure deployment, security compliance, and storage optimization**.
- **Built and optimized a secure, scalable data lake** integrating structured and unstructured financial data, enhancing data accuracy and security.
- **Automated ETL workflows** with **Apache Airflow**, improving data ingestion, transformation, and fraud risk assessment processing efficiency.
- **Reduced transaction processing latency** by **enhancing Spark-based streaming workflows**, enabling real-time fraud alerts.
- **Implemented PCI-DSS-compliant data governance**, including **data encryption, lineage tracking, and access control** across multiple cloud-based services.
- **Developed fraud analytics dashboards using Power BI**, integrating real-time transaction anomaly detection and risk scoring to support fraud prevention strategies.

- **Collaborated with Risk, Compliance, and Data Science teams** to design and optimize machine learning models for fraud detection.
- **Optimized AWS S3 lifecycle policies and data partitioning**, ensuring **faster retrieval for fraud detection models**.

Humana Healthcare Inc, Louisville, KY

January 2023 – March 2024

Data Engineer

Responsibilities:

- **Built scalable ETL pipelines for patient and provider performance analytics** using **Python, SQL, AWS Redshift, and Glue**, ensuring efficient data processing.
- **Developed predictive analytics models** in collaboration with **Data Science teams**, enabling risk stratification and provider ranking based on performance metrics.
- **Developed real-time analytics pipelines for patient monitoring and claims processing**, enhancing decision-making for healthcare providers and reducing approval times.
- **Ensured HIPAA-compliant data security** by implementing **role-based access controls, AWS IAM policies, encryption, and automated auditing within AWS-based systems**.
- **Reduced data processing time** by optimizing **Spark-based batch jobs**, accelerating claims processing and healthcare analytics.
- **Designed and automated ETL workflow execution using Apache Airflow and AWS Step Functions**, reducing failures, improving scalability, and leveraging **Terraform for infrastructure automation**.
- **Developed Power BI dashboards**, providing **real-time insights into patient outcomes and provider performance metrics**.
- **Optimized cloud storage costs** through intelligent data partitioning and lifecycle management for healthcare data.

Reliance Retail, Bengaluru, India

January 2021 – Nov 2022

Data Analyst

Responsibilities:

- **Developed a real-time inventory tracking system** using **Apache Kafka, Spark Streaming, and AWS S3**, ensuring accurate stock level monitoring across multiple warehouses and retail locations.
- **Built and optimized ETL pipelines** to process and integrate data from **POS systems, supply chain sources, and warehouse databases**, ensuring seamless data synchronization and accuracy.
- **Designed demand forecasting models** leveraging **predictive analytics and machine learning**, improving stock availability and reducing supply chain inefficiencies.
- **Developed and optimized ETL workflows using AWS Glue**, automating **inventory data ingestion and improving supply chain data consistency across warehouses and POS systems**.
- **Developed Power BI dashboards**, providing **real-time visibility into stock levels, sales performance, and demand trends for business users and operations teams**.
- **Implemented data security and compliance measures**, enforcing **access controls and monitoring solutions** to protect sensitive product and customer data.
- Collaborated with cross-functional teams, including supply chain, operations, and business intelligence, to define key inventory metrics, enhance reporting accuracy and drive data-driven decision making.

Education

- **Masters in Computer Science**
- Lindsey Wilson College, Columbia, KY.